

ANAND PINNAMANENI

+1 857-351-5245 | pinnamanenianand177@gmail.com | LinkedIn: anandp17 | GitHub: Anand283

EDUCATION

Northeastern University, Khoury College of Computer Sciences

Master of Science in Computer Science

Boston, MA

Jan 2025 – Dec 2026

Koneru Lakshmaiah Education Foundation

Bachelor of Technology in Computer Science

Vijayawada, India

Oct 2019 – May 2023

TECHNICAL PROJECTS

CPU Ray Tracer | C++17, SDL3, CMake

Apr 2026 – May 2026

- Built a from-scratch **Monte Carlo ray tracer** implementing the full physical light-transport pipeline with Lambertian, fuzzed metallic, and **dielectric** materials (Snell's law refraction, **Schlick's Fresnel**, total internal reflection), plus jittered anti-aliasing and depth-of-field via aperture sampling.
- Designed the camera model from first principles: look-from / look-at / up basis vectors, viewport plane construction, and a defocus disk, internalizing the projective geometry shared with multi-view reconstruction.
- Rendered a 485-sphere scene at 1200×675 with 100 samples/pixel and 50-bounce recursion, displayed live through an SDL3 streaming texture; verified visually against the *Ray Tracing in One Weekend* reference for Fresnel rings, soft shadows, and focal-plane sharpness.

Distributed Sharded Key-Value Store | C++, Sockets, Multithreading, WAL

Sep 2025 – Dec 2025

- Built a sharded, fault-tolerant key-value store sustaining **9,000–11,000 ops/sec** with linear horizontal scaling across 1, 3, and 5 shards on a 90% read workload.
- Engineered the load balancer with **Rendezvous Hashing (HRW)** for stable replica selection and TCP health probes, sustaining ≈0.9% throughput degradation during a live shard failure.
- Implemented a **write-ahead log** with length-prefixed binary encoding and write-before-modify ordering, validating **100% key recovery** via WAL replay after deliberate shard crash and restart.
- Designed a compact binary KV protocol (12-byte request, 8-byte response headers) for safe, fixed-cost framing over POSIX TCP sockets.

StudyGroupApp Backend | Node.js, Express, Socket.io, Google OAuth, Supabase, PostgreSQL

Jan 2025 – Apr 2025

- Built the room-presence layer syncing an active_room_users Supabase table with **Socket.io** in-memory rooms, propagating join/leave events in real time and preventing ghost presences on client navigation.
- Implemented dual-layer Google OAuth with Express middleware for REST routes and a Socket.io handshake middleware, attaching a pre-validated user to every socket and eliminating per-event token re-verification.
- Designed a get-or-create document API enforcing one active Etherpad / Excalidraw session per room per editor type, with deterministic IDs guaranteeing idempotent URL resolution across clients.

EXPERIENCE

Salesforce

Associate Technical Support Engineer

Hyderabad, India

Aug 2023 – Dec 2024

- Operated a 15-minute production-incident SLA on Salesforce's multi-tenant managed platform, triaging 20 to 30 daily alerts from an ML-driven detector and root-causing concurrency failures across REST/SOAP APIs, async message queues, pub/sub, and the database layer.
- Diagnosed row-lock contention, ACID violations, governor-limit breaches, and chunking failures by tracing request flows through distributed system layers and writing **Splunk** queries against production telemetry.

Koneru Lakshmaiah Education Foundation

Student Peer Mentor

Vijayawada, India

July 2020 – Dec 2020

- Mentored 20+ students in AI fundamentals (search algorithms, fuzzy logic, neural networks, knowledge representation) and co-designed 12 lab experiments with faculty for the Data Structures course (cohort of 150+).

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, C#, Java, SQL

Systems / Backend: POSIX sockets, multithreading, write-ahead logging, Node.js, Express, Socket.io, PostgreSQL, REST, OAuth

Graphics / CV: SDL3, OpenCV, PyTorch, NumPy, GTSAM, Open3D, Unity, CMake

Concepts: Distributed systems, replication, consistent hashing, projective geometry, Monte Carlo ray tracing, real-time rendering

Relevant Coursework: Pattern Recognition & Computer Vision (CS 5330), Algorithms (CS 5800), Building Scalable Distributed Systems (CS 6650)